Om Shedo
THADOMAL SHAHANI
TSEC
ENGINEERING COLLEGE
BE C31
2103163

Experiment 10

Aim: Case study: CORBA, Android stack

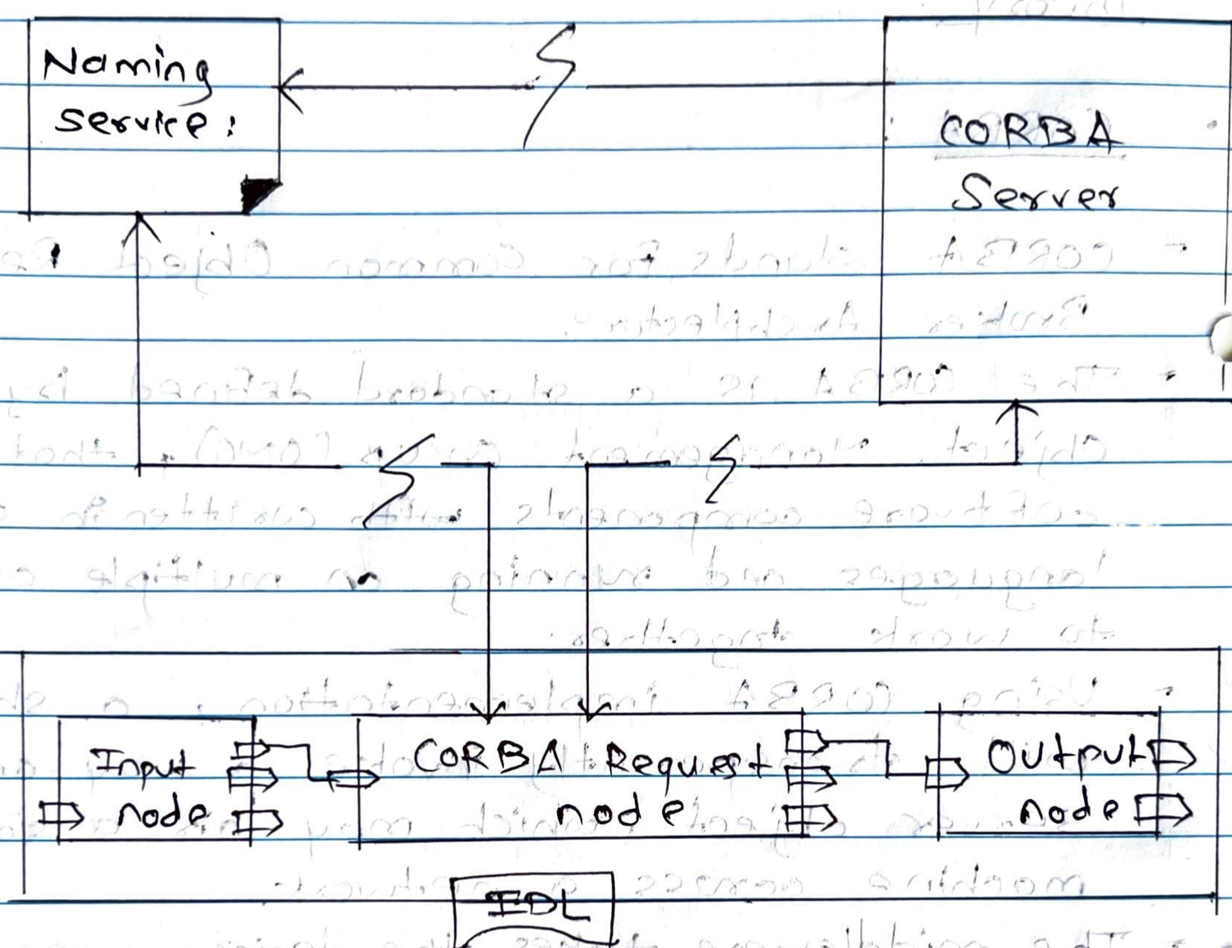
Theory:

- **CORBA**:

- CORBA stands for Common Object Request Broker Architecture.
- The CORBA is a standard defined by the Object Management Group (OMG), that enables software components ~~with~~ written in computer languages and running on multiple computers to work together.
- Using CORBA implementation, a shopper will transparently invoke a way on a server object, which may air a simillar machine across a network.
- The middleware takes the decision, associated is to blame for finding an object that will implement the request, passing its parameters, invoking its methodology, and returning the results of the invocation.
- CORBA is not associated with a particular programming language and any language with a CORBA binding can be used to call and implement objects.
- Objects are defined in a syntax called ~~Integer~~ Interface Definition Language (IDL).

THADOMAL SHAHANI
TSEC
ENGINEERING COLLEGE

→ CORBA includes four components :-



1) ORB : It handles the communication, marshaling and unmarshaling of parameters so that the parameter handling is transparent for a CORBA server and client applications.

2) CORBA Server : It creates objects and initialize them with an ORB. The server places references to the CORBA objects inside a naming service so that clients can access them.

3) Naming service : It holds references to CORBA objects.

4) CORBA Request Node : It acts as a client.

- Android stack

- The Android software stack generally consists of Linux kernel and a collection of C/C++ libraries that are exposed through an application framework, that provides services and management of the apps and run time.
- Android was created on the open-source kernel of Linux.
- Features :-

1) Security : It handles the security between the application and system.
2) Memory Management : It efficiently handles memory management thereby providing the freedom to develop our apps.
3) Network stack : It effectively handles network communication.
4) Driver model : It ensures the apps works.

- Libraries :

Running on top of kernel, the framework was developed with features.

THADOMAL SHAHANI
TSEC
ENGINEERING COLLEGE

1) Android runtime : It consists of core libraries of Java and ART.
2) Open GL : This cross-language is used to produce 2D and 3D graphics.
3) Webkit : This open-source web browser engine provides all the functionality to display contend.
4) Media framework : They allow you to play and record audio & video.
5) Secure Socket layer (SSL) : They are there for Internet security.

- Applications :

It can be found at the topmost layer. At the application layer, we write our application to be insulted on this layer only.
E.g., are games, messages, contacts, etc.